

OmniTrade Hub

DATABASE SCHEMA

GROUP 10

1. Users Table (Handles Customers, Vendors, and Administrators)

```
CREATE TABLE Users (  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  Username VARCHAR(100) NOT NULL UNIQUE,  
  Email VARCHAR(150) NOT NULL UNIQUE,  
  PasswordHash VARCHAR(255) NOT NULL,  
  Role VARCHAR(50) NOT NULL CHECK (Role IN ('Customer', 'Vendor',  
  'Admin')),  
  CreatedAt DATETIME DEFAULT GETDATE()  
);
```

2. Vendors Table (Storefront profile for merchants)

```
CREATE TABLE Vendors (  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  UserId INT NOT NULL,  
  StoreName VARCHAR(150) NOT NULL,  
  Description VARCHAR(MAX),  
  ContactEmail VARCHAR(150) NOT NULL,  
  IsApproved BIT DEFAULT 0,  
  CONSTRAINT FK_Vendor_User FOREIGN KEY (UserId) REFERENCES Users(Id)  
  ON DELETE CASCADE  
);
```

3. Products Table (Product catalog managed by vendors)

```
CREATE TABLE Products (  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  VendorId INT NOT NULL,  
  Name VARCHAR(150) NOT NULL,  
  Description VARCHAR(MAX),  
  Price DECIMAL(18, 2) NOT NULL CHECK (Price >= 0),  
  StockQuantity INT NOT NULL CHECK (StockQuantity >= 0),  
  Category VARCHAR(100) NOT NULL,  
  CONSTRAINT FK_Product_Vendor FOREIGN KEY (VendorId) REFERENCES  
  Vendors(Id) ON DELETE CASCADE  
);
```

4. CartItems Table (Temporary shopping cart storage for buyers)

```
CREATE TABLE CartItems (  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  CustomerId INT NOT NULL,  
  ProductId INT NOT NULL,  
  Quantity INT NOT NULL CHECK (Quantity > 0),  
  CONSTRAINT FK_CartItem_Customer FOREIGN KEY (CustomerId) REFERENCES  
  Users(Id) ON DELETE CASCADE,  
  CONSTRAINT FK_CartItem_Product FOREIGN KEY (ProductId) REFERENCES  
  Products(Id) ON DELETE CASCADE  
);
```

5. Orders Table (Master order entity generated during checkout)

```
CREATE TABLE Orders (  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  CustomerId INT NOT NULL,  
  OrderDate DATETIME DEFAULT GETDATE(),  
  TotalAmount DECIMAL(18, 2) NOT NULL CHECK (TotalAmount >= 0),  
  Status VARCHAR(50) NOT NULL DEFAULT 'Pending' CHECK (Status IN  
    ('Pending', 'Processing', 'Shipped', 'Delivered', 'Refunded')),  
  CONSTRAINT FK_Order_Customer FOREIGN KEY (CustomerId) REFERENCES  
    Users(Id)  
);
```

6. OrderItems Table (Line items mapped to specific vendors)

```
CREATE TABLE OrderItems (  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  OrderId INT NOT NULL,  
  ProductId INT NOT NULL,  
  VendorId INT NOT NULL,  
  Quantity INT NOT NULL CHECK (Quantity > 0),  
  UnitPrice DECIMAL(18, 2) NOT NULL CHECK (UnitPrice >= 0),  
  CONSTRAINT FK_OrderItem_Order FOREIGN KEY (OrderId) REFERENCES  
    Orders(Id) ON DELETE CASCADE,  
  CONSTRAINT FK_OrderItem_Product FOREIGN KEY (ProductId) REFERENCES  
    Products(Id),  
  CONSTRAINT FK_OrderItem_Vendor FOREIGN KEY (VendorId) REFERENCES  
    Vendors(Id)  
);
```